

A crossed random-effects latent class model for annotation ensembles of large language models

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Abstract

Large language models (LLMs) are increasingly used to label text at scale, but every label is produced by a *configuration*—a model, a prompt, and a stochastic decoding run—and configurations that share training corpora and alignment procedures do not err independently. Classical aggregation models for multiple raters, from Dawid and Skene onward, assume conditional independence of raters given the true class. We show by simulation that, under the crossed dependence typical of LLM ensembles, such models increasingly overstate annotator accuracy and bias prevalence estimates as dependence grows. We propose a latent class model in which the dependence structure is derived from the factorial design of the annotation ensemble: item-level random effects that are shared across all configurations, crossed with item-by-model and item-by-prompt effects, while repeated decoding runs are treated as exchangeable. The model, estimated by Bayesian inference with the latent class marginalised, recovers point identification without a gold standard, decomposes annotation error variance into item, model, prompt and run components, and yields calibrated posterior class probabilities. We study identifiability, evaluate the method against majority vote, Dawid–Skene and partial-identification bounds in a simulation study, and apply it to policy-topic coding of legislative texts with human-coded reference labels. The variance decomposition translates directly into design guidance for how many models, prompts and runs to use.

Keywords: latent class analysis, conditional dependence, random effects, large language models, text annotation, measurement error

MSC Classification: 62H30 , 62F15 , 62J12 , 62P25 , 68T50

1 Introduction

Text classification by large language models has moved, within three years, from a curiosity to routine research infrastructure. Studies in the social sciences, management and public administration now use LLMs to label sentiment, policy topics, stance, toxicity or compliance in corpora far too large for manual coding (Gilardi et al. 2023; Carlson and Burbano 2026). The attraction is obvious: labels arrive at a fraction of the cost and time of human annotation, and early comparisons suggested accuracy on a par with crowd workers.

The reliability of those labels is a different matter. An LLM label is not the output of a single, stable instrument. It is the output of a configuration: a particular model, a particular prompt wording, and a particular stochastic decoding run. Each of these varies, and each variation moves the label. Semantically equivalent prompts on the same task can shift accuracy by more than twenty points (Liu 2026); nominally deterministic decoding settings do not in fact return deterministic output (Atil et al. 2024); and re-running the same pipeline months later yields different labels because the underlying model has changed (Barrie et al. 2025). At scale, these choices have downstream consequences. Baumann et al. (2025) replicated 37 annotation tasks from 21 published studies across 18 models and found that the choice of model, prompt and temperature alone was enough to change the sign or significance of substantive conclusions—a phenomenon they call LLM hacking. Camuffo et al. (2026) report that in their strategy-research setting the bulk of annotation variance was attributable to which model was queried rather than to the content being labelled.

The natural statistical response is to treat each configuration as a rater and aggregate. This is precisely what recent methodological guidance recommends: query several models with several prompts, replicate, and combine the labels with an aggregation model such as Dawid and Skene (1979) or GLAD (Whitehill et al. 2009) that estimates each rater’s error rates jointly with the latent true labels (Camuffo et al. 2026). The recommendation is sensible, but it imports an assumption that the LLM setting violates in a specific and predictable way. Dawid–Skene and its descendants assume that raters are conditionally independent given the true class. Human crowd workers approximate this reasonably well; they err for idiosyncratic reasons. LLM configurations do not. Two prompts sent to the same model share that model’s training data, tokenisation and alignment; two models given the same prompt share the prompt’s framing and any ambiguity in the item itself; and all configurations share whatever makes a particular text genuinely hard to classify. The errors are therefore correlated, and the correlation has structure.

The diagnostic-testing literature met exactly this problem thirty years ago, when multiple imperfect tests applied to the same patient were found to share unmeasured patient-level characteristics. The conditional-independence latent class model fits such data poorly and biases estimated sensitivities and specificities (Vacek 1985). Qu et al. (1996) introduced subject-level random effects into the latent class model to absorb the shared dependence, and the approach has since been extended, tested and implemented in software (Oberski et al. 2013; Beath 2017; Xu et al. 2019). Within the classification community, relaxations of local independence have been developed for model-based clustering of categorical data (Marbac et al. 2016; Oberski 2016). What these models

do not do is exploit the fact that, in an LLM ensemble, the rater is itself a designed object with a known factorial structure.

This paper makes three contributions. First, we propose a latent class model—the crossed random-effects latent class model, CRE-LCM—in which the dependence between LLM configurations is neither ignored nor left unrestricted but derived from the design of the annotation ensemble: an item effect shared by all configurations, crossed with item-by-model and item-by-prompt effects, with repeated decoding runs treated as exchangeable given these effects. Dawid–Skene and the one-factor model of [Qu et al. \(1996\)](#) are nested special cases. Second, we study identifiability. The crossed effects are identified by the replication structure of the design; the global item effect competes with the latent class itself, and we characterise the conditions and the remedies under which point identification is recovered, in contrast to the partial-identification route recently proposed for the same problem ([Chen et al. 2026](#)). Third, we show through simulation that independence-based aggregation overstates accuracy and biases prevalence monotonically in the strength of dependence, that CRE-LCM removes this bias, and that its variance decomposition provides direct guidance for annotation design: whether to add models, prompts or runs. An application to policy-topic coding of legislative texts, with human reference labels held out for validation, illustrates the method.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the model, its special cases, identifiability and estimation. Section 4 reports the simulation study. Section 5 presents the application. Section 6 discusses design implications and limitations.

2 Background and related work

2.1 Aggregating labels from multiple raters without a gold standard

The joint estimation of true labels and rater error rates was formalised by [Dawid and Skene \(1979\)](#), who modelled each rater by a class-conditional confusion matrix and used EM. The model remains the reference point for crowdsourced labelling and unsupervised ensemble aggregation. Extensions add item difficulty (GLAD; [Whitehill et al. 2009](#)), Bayesian priors and hierarchical structure for sparse-label regimes, spectral initialisation, and online estimation. Within the classification literature, [Salter-Townshend and Murphy \(2014\)](#) fitted a finite mixture to annotator bias parameters in order to cluster inexpert sentiment annotators, showing that annotator heterogeneity is itself a clustering problem; [Raykar and Yu \(2012\)](#) ranked and filtered crowd annotators by estimated reliability. Agreement statistics such as multi-rater kappas ([Warrens 2010, 2013](#)) summarise concordance descriptively but do not estimate accuracy against a latent truth. The common thread in the model-based approaches is conditional independence of raters given the true class. The current LLM-annotation guidance—treat each model or model–prompt pair as an annotator and apply Dawid–Skene or GLAD via standard packages ([Camuffo et al. 2026](#))—inherits this assumption.

2.2 Conditional dependence in latent class models

Violations of local independence are among the oldest concerns in latent class analysis. In diagnostic testing, Vacek (1985) showed that ignoring dependence between tests biases estimated accuracies, and Qu et al. (1996) introduced a latent class model with subject-level random effects, including a class-specific scale, to absorb it. The literature since has developed detection methods—bivariate residuals, bootstrapped residuals and score tests, whose relative performance was evaluated in this journal by Oberski et al. (2013)—and alternative parameterisations: log-linear (fixed pairwise) versus Gaussian random-effects (GRE) formulations, with GRE more parsimonious when dependence is pervasive but numerically more delicate (Xu et al. 2019). Crossed subject-by-rater random effects have been used for ordinal ratings without a gold standard, with robustness checks against misspecified random-effect distributions (Xie et al. 2013). Oberski (2016) distinguished substantive from nuisance dependence, modelling the former as additional discrete latent variables and absorbing the latter in extra parameters, with an application to misclassification of self-reported voting. In model-based clustering of categorical data, Marbac et al. (2016) relaxed local independence by grouping variables into conditionally independent blocks with parsimonious intra-block dependence. Software is available (randomLCA; Beath 2017).

None of this work treats the rater as a factorial object. The rater is a single unit—a test, a pathologist, a survey item—and dependence is either estimated freely or absorbed into a single subject-level factor. In an LLM ensemble the rater has a known design, model $\times$ prompt $\times$ run, and the design tells us where dependence can arise. This is the gap we address.

2.3 LLM annotation: variability, measurement error, and remedies

Three strands of recent work bear on the problem. The first documents instability: accuracy varies substantially across meaning-preserving prompt paraphrases (Liu 2026; Barrie et al. 2024), across decoding runs (Atil et al. 2024), and across time as models are updated (Barrie et al. 2025); large-scale audits of LLM judges find high inter-model agreement but limited validity against task-specific human annotation (Bavaresco et al. 2025; Norman et al. 2026). The second traces consequences downstream: Baumann et al. (2025) quantified how configuration choices propagate into Type I, II, S and M errors in 2,361 hypothesis tests; Egami et al. (2024) and Camuffo et al. (2026) show that annotation error correlated with covariates yields inconsistent regression estimates regardless of average accuracy; Messing (2026) extends the argument to evaluation and benchmarking.

The third strand proposes statistical remedies, and two are closest to ours. Zhang and Martinez (2025) reframe LLM stochasticity as measurement error and propose a Bayesian generalised-linear version of Dawid–Skene with covariates on the latent state; their focus is repeated draws from a single pipeline, and raters remain conditionally independent. Chen et al. (2026) takes the opposite route: because LLMs share corpora and alignment, their errors may be arbitrarily dependent, so the authors

abandon point identification and derive bounds on prevalence, tightened by reporter-specific calibration. Prediction-powered inference (Angelopoulos et al. 2023) offers a design-based alternative that requires a labelled subsample. Protocol papers (Camuffo et al. 2026; Xu et al. 2026) recommend prompt ensembles, replication and noise-aware aggregation but supply no model in which the dependence between configurations is estimated.

2.4 Positioning

The literature therefore offers independence-based aggregators known to be mis-specified for LLM ensembles; bounds that discard the information contained in the ensemble’s design; and protocols without an accompanying inferential model. CRE-LCM occupies the middle ground. It gives dependence the structure implied by how the annotations were generated, which yields point identification under stated conditions, interpretable variance components that translate directly into annotation-design decisions, and a natural home for the covariate and multiclass extensions already present in the existing approaches. Methodologically it sits in the ADAC tradition of relaxing local independence in latent class and mixture models (Oberski et al. 2013; Marbac et al. 2016; Oberski 2016) and of modelling annotator heterogeneity as a statistical object (Salter-Townshend and Murphy 2014).

3 Model

3.1 Setting and notation

Items $i = 1, \dots, N$ have an unobserved binary class $c_i \in \{0, 1\}$ (K classes in Section 3.5) with prevalence $\pi_1 = \Pr(c_i = 1)$. Each item is annotated by every configuration (m, p, r) of M models, P prompts and R decoding runs, giving binary labels Y_{impr} . Runs are exchangeable within a model-prompt pair, so the sufficient statistic is the count $S_{imp} = \sum_r Y_{impr} \in \{0, \dots, R\}$.

3.2 The crossed random-effects latent class model

Conditional on the class and on three item-level random effects, the labels are independent Bernoulli variables with

$$\text{logit } \Pr(Y_{impr} = 1 \mid c_i = k, \theta_i, \phi_{im}, \psi_{ip}) = \mu_k + a_{mk} + b_{pk} + \theta_i + \phi_{im} + \psi_{ip}, \quad (1)$$

where $\mu_0 < \mu_1$ are ordered class intercepts, a_{mk} are fixed model effects with the reference model $a_{1k} = 0$, $b_{pk} \sim N(0, \tau_k^2)$ are random prompt effects, and

$$\theta_i \sim N(0, \sigma_\theta^2), \quad \phi_{im} \sim N(0, \sigma_\phi^2), \quad \psi_{ip} \sim N(0, \sigma_\psi^2) \quad (2)$$

are mutually independent: θ_i is the ambiguity of item i shared by all configurations, ϕ_{im} the item×model effect and ψ_{ip} the item×prompt effect. Runs are conditionally independent given these effects, so $S_{imp} \mid c_i = k, \theta_i, \phi_{im}, \psi_{ip} \sim$

$\text{Binomial}(R, \text{expit}(\eta_{impk}))$ with η_{impk} the right-hand side of (1), and the marginal likelihood of item i is

$$L_i = \sum_{k=0}^1 \pi_k \int \prod_{m=1}^M \prod_{p=1}^P \binom{R}{S_{imp}} \text{expit}(\eta_{impk})^{S_{imp}} (1 - \text{expit}(\eta_{impk}))^{R - S_{imp}} dF(\theta_i, \phi_i, \psi_i). \quad (3)$$

Nested cases: with $\theta = \phi = \psi = 0$ the model is Dawid–Skene with MP raters whose sensitivities and specificities are $\text{expit}(\mu_1 + a_{m1} + b_{p1})$ and $1 - \text{expit}(\mu_0 + a_{m0} + b_{p0})$; with $\phi = \psi = 0$ it is the one-factor random-effects model of Qu et al. (1996) with a common loading. The random effects play the role of the nuisance-dependence parameters of Oberski (2016), with the difference that their placement is dictated by the design.

3.3 Variance decomposition

On the latent logistic scale the variance of the label propensity of an item over configurations and runs is $\sigma_\theta^2 + \sigma_\phi^2 + \sigma_\psi^2 + \pi^2/3$, the last term being the variance of the standard logistic distribution that generates the run-level Bernoulli outcome. The shares

$$s_{\text{item}} = \frac{\sigma_\theta^2}{V}, \quad s_{\text{model}} = \frac{\sigma_\phi^2}{V}, \quad s_{\text{prompt}} = \frac{\sigma_\psi^2}{V}, \quad s_{\text{run}} = \frac{\pi^2/3}{V}, \quad V = \sigma_\theta^2 + \sigma_\phi^2 + \sigma_\psi^2 + \pi^2/3, \quad (4)$$

have a direct design interpretation: adding models reduces the ϕ share, adding prompts reduces the ψ share, adding runs reduces only the $\pi^2/3$ share, and θ is irreducible by design.

3.4 Identifiability and the merged-class mode

Two questions arise. Are the parameters of CRE-LCM identified without a gold standard, and can they be estimated in practice? We address both with the scenario-B design of Section 4 ($N = 300$, $M = 3$, $P = 5$, $R = 3$; $\sigma_\theta = 1.0$, $\sigma_\phi = 0.8$, $\sigma_\psi = 0.5$), fitted with four chains of 2,000 iterations.

Formal identification. The crossed effects ϕ_{im} and ψ_{ip} are identified by the replication structure: each item is observed under P prompts within every model and under M models within every prompt, so item-by-model and item-by-prompt deviations are separable from the run-level binomial noise whenever $M \geq 2$, $P \geq 2$ and $R \geq 2$. The class intercepts are identified up to label permutation, which we resolve by ordering $\mu_0 < \mu_1$; the model effects are identified relative to a reference model. The condition of Nguyen et al. (2023)—at least $2K - 1$ conditionally exchangeable noisy labels per item for a K -class mixture—is met with room to spare ($MPR = 45$ labels per item for $K = 2$), and the fixed-effect part of the model is a Dawid–Skene model with a shared logistic random effect, whose identification for $M \geq 3$ raters follows Jones et al. (2010). The delicate parameter is σ_θ : an item that every configuration labels 1 can be explained either as an item of class 1 or as an ambiguous item with a large θ_i , so item ambiguity and class membership are partially confounded. This is the same mechanism that Qu et al. (1996) describe for the shared subject effect in diagnostic testing.

The merged-class mode. In practice the more consequential problem was not weak identification but multimodality. With Stan’s default random initialisation, one chain in four converged to a mode in which the mixture has collapsed: prevalence near 0.70, class intercepts almost equal (both between -0.4 and 0), model effects with reversed signs and inflated prompt variances, all heterogeneity being carried by the item effects. The remaining chains agreed with each other and with the truth. Pooling across chains—as a naive analysis would—yields a prevalence of 0.42 [0.27, 0.74] for a true value of 0.30 and an $\hat{R}$ of 1.81; the estimate is an average of two modes. This mode is distinct from label switching (the classes do not exchange; they merge) and is not removed by standard relabelling.

Remedies compared. Table 2 summarises five remedies. Two that seem natural do not work. A class-specific scale for θ_i , in the spirit of Qu et al. (1996), makes matters worse: with $M = 3$ the extra scale trades off against the mixing proportion and the four chains stop at prevalences of 0.17, 0.23, 0.33 and 0.51, the highest-density chain being the one at 0.17. Dropping θ_i altogether—using only the crossed effects—leaves the merged mode in place and pushes the shared item variance into the item-by-model component, whose SD rises from 0.80 to 1.06 [0.97, 1.16] and takes every variance share off target. A tighter prior on σ_θ changes nothing; the data dominate it. Two remedies work. Anchoring 5% of items (15 of 300) with known labels removes the merged mode entirely: all chains agree, all intervals cover the truth, prevalence 0.31 [0.22, 0.38], σ_θ 0.93 [0.77, 1.12], and the posterior labels are the best of any variant (accuracy 0.913, Brier 0.063 over all items; 0.909 and 0.067 on the 285 non-anchored items). Initialising the chains at the Dawid–Skene solution, without changing the model, also removes it: prevalence 0.32 [0.26, 0.39], intercepts $(-2.08, 1.20)$ for $(-2.00, 1.50)$, model effects recovered, accuracy 0.913, Brier 0.084, in 2.4 minutes against 10 for random starts. The base model is therefore estimable without any gold standard once the sampler starts near the data-consistent mode; we carry this version forward and report anchoring as the practical remedy for users who hold a small labelled subset—the situation of Section 5.

A residual bias. In every chain that finds the correct mode, whatever the variant, σ_θ is shrunk by 10–20% (0.79–0.92 for a true 1.00; 0.83 [0.67, 1.00] with Dawid–Skene initialisation). Anchoring reduces the shrinkage to 7%. This is the confounding of item ambiguity with class membership noted above and is a property of the likelihood at this design size, not of the estimator; the item share of variance is correspondingly under-estimated (0.14 for 0.19) while the model, prompt and run shares are recovered. We report it as a limitation and examine its dependence on N , P and the strength of dependence in Section 4.

Model comparison. Leave-one-out cross-validation is not usable here: each item carries $1 + M + P$ random effects of its own, so removing an item changes the posterior substantially ($p_{\text{loo}} \approx 750$ for $N = 300$; 88–93% of Pareto- k values above 0.7 in every variant). Comparisons between variants therefore rest on parameter recovery in simulation, not on predictive criteria.

3.5 Multiclass extension

[To be decided after the simulation study.]

3.6 Estimation

We marginalise the discrete class and sample the continuous parameters with the no-U-turn sampler (Stan via cmdstanr; [Carpenter et al. 2017](#)), using a non-centred parameterisation for all random effects, weakly informative priors ($\mu_k \sim N(0, 2)$ ordered; model effects $N(0, 1)$; half-normal(0, 1) on the three SDs; half-normal(0, 0.5) on the prompt-effect SDs τ_k ; Beta(1, 1) on the prevalence), and adapt_delta = 0.9. Chains are initialised at the Dawid–Skene EM solution for the class prevalence and intercepts, with random effects at zero; convergence is assessed per chain with split- $\hat{R}$ and bulk and tail effective sample sizes ([Vehtari et al. 2021](#)), and a merged-class mode is diagnosed by a chain whose prevalence and intercepts depart from the others. Posterior class probabilities $\Pr(c_i = 1 \mid \text{data})$ and the variance shares are computed in generated quantities. Mixing on the prevalence and σ_θ is slow (bulk ESS of roughly 13–15 per 4,000 draws at $N = 300$), so we recommend at least 1,500 iterations per chain for these two parameters; all other parameters mix well. Runtime is about 70 seconds per chain per 1,000 iterations at $N = 300$ on one core. An EM algorithm with adaptive quadrature over the item effects would be a faster alternative and is left for future work. The leave-one-out diagnostics of Section 3.4 use Pareto-smoothed importance sampling ([Vehtari et al. 2017](#)). The Stan programs for the main model and the variants of Section 3.4 are listed in Appendix A.

4 Simulation study

4.1 Design

Data are generated from (1)–(2) with $M = 3$ models, $R = 3$ runs, $\pi_1 = 0.30$, $\mu = (-2, 1.5)$, model effects $a_{\cdot 0} = (0, 0.3, 0.6)$ and $a_{\cdot 1} = (0, -0.1, -0.5)$ (so that later models have both a higher false-positive rate and a lower sensitivity), prompt effects with $\tau_0 = \tau_1 = 0.4$, and random-effect standard deviations $(\sigma_\theta, \sigma_\phi, \sigma_\psi) = \delta \times (1.0, 0.8, 0.5)$. The dependence level $\delta \in \{0, 0.5, 1, 1.5\}$ is crossed with $N \in \{150, 300, 600\}$ and $P \in \{3, 5\}$, giving 24 cells with 100 replications each. A misspecification arm at $\delta = 1$, $N = 300$, $P = 5$ replaces the three Gaussian random effects by scaled Student- t_4 variables with the same standard deviations. Every replication has a recorded seed (data and sampler) and is stored individually, so the study is reproducible and resumable from the public code repository.

Three estimators are compared: majority vote over the *MPR* labels; the binary Dawid–Skene model with *MPR* raters fitted by EM ([Dawid and Skene 1979](#)); and CRE-LCM as described in Section 3.6, with two chains of 1500 iterations (750 warm-up) per replication, initialised at the Dawid–Skene solution. For each replication we record the bias of the estimated prevalence, of mean sensitivity and of mean specificity (averaged over the $M \times P$ configurations; for CRE-LCM these are the posterior means of the marginal accuracies implied by the model, integrating over the random effects), classification accuracy and Brier score of the posterior class probabilities against the simulated classes, coverage of the CRE-LCM 95% credible intervals, and the estimated variance shares (4). Replications in which any global parameter of CRE-LCM has split- $\hat{R} > 1.05$ are flagged and their fraction reported per cell.

4.2 Dawid–Skene under dependence

Table 1 gives a first view on a single data set per dependence level. Under conditional independence ($\delta = 0$) Dawid–Skene recovers sensitivity, specificity and prevalence essentially exactly. As δ grows the true marginal accuracies fall, because the item effects push labels away from the class-specific propensities, but the Dawid–Skene estimates do not: at $\delta = 1.5$ the model reports a specificity of 0.82 against a true 0.74 and a prevalence of 0.36 against 0.29. The bias is monotone in δ and is larger for prevalence and specificity than for sensitivity, which the results over the full grid below confirm.

Table 1 Dawid–Skene under crossed dependence: one simulated data set per dependence level ($N = 300$, $M = 3$, $P = 5$, $R = 3$, seed 2026). Dependence level δ scales the three random-effect standard deviations $(\sigma_\theta, \sigma_\phi, \sigma_\psi) = \delta \times (1.0, 0.8, 0.5)$. Sensitivity and specificity are averaged over the $M \times P$ configurations

δ	Sensitivity		Specificity		Prevalence		
	true	DS	true	DS	true	DS	MV
0.0	0.710	0.710	0.853	0.856	0.290	0.290	0.290
0.5	0.696	0.720	0.832	0.832	0.290	0.287	0.280
1.0	0.666	0.703	0.787	0.822	0.290	0.327	0.287
1.5	0.638	0.714	0.741	0.819	0.290	0.359	0.307

4.3 Identifiability variants

Table 2 reports the five remedies of Section 3.4, fitted to one scenario at $\delta = 1$, $N = 300$, $P = 5$ with four chains of 2000 iterations. With default random initialisation, one of the four chains of the base model settles in the merged-class mode (per-chain prevalence near 0.70, class intercepts almost equal, both between -0.4 and 0 , model effects sign-flipped); the pooled posterior is a mixture of the two modes and $\hat{R}$ on π_1 is 1.81. The class-specific scale (a) makes matters worse: each chain ends at a different prevalence (0.17 to 0.51) and the chain with the highest posterior density is the one furthest from the truth, so the extra loading is not identified separately from the mixing proportion in this design. Omitting θ_i (b) does not remove the mode and biases the remaining components, the item $\times$ model standard deviation rising to 1.06 against 0.80, because the shared item variance is pushed into the crossed effects. The informative prior (d) changes nothing, the data dominating the prior. Anchoring 5% of the items (c) removes the mode entirely, with every interval covering the truth and the best label quality of all variants. Initialising the chains at the Dawid–Skene solution (e) also removes it: all chains agree, the intervals cover the truth, and label quality is close to that of anchoring. Two limitations remain under (e): mixing on π_1 and σ_θ is slow, and σ_θ is shrunk downwards by 10–20% in every chain that sits in the correct mode of every variant, anchoring reducing the shrinkage. We therefore carry

variant (e) forward as the main model and report anchoring as the practical remedy when a small labelled subset exists.

Table 2 Identifiability variants fitted to one scenario-B data set ($N = 300$, $\sigma_\theta = 1$, $\sigma_\phi = 0.8$, $\sigma_\psi = 0.5$, true $\pi_1 = 0.30$, $\mu = (-2, 1.5)$; four chains of 2000 iterations). Posterior means, with 95% credible intervals for π_1 and σ_θ ; $\hat{R}$ is the largest split- $\hat{R}$ over the global parameters

Variant	π_1	μ_0, μ_1	σ_θ	Acc.	Brier	max $\hat{R}$
Majority vote	—	—	—	0.890	0.110	—
Dawid–Skene	—	—	—	0.887	0.109	—
Base	0.42 [0.27, 0.74]	-1.84, 0.65	0.86 [0.69, 1.09]	0.897	0.121	1.81
(a) class-specific σ_θ	0.31 [0.14, 0.60]	-1.52, 1.15	1.17 / 0.80	0.897	0.107	2.67
(b) θ omitted	0.41 [0.28, 0.71]	-1.62, 0.92	0.00 [0.00, 0.00]	0.897	0.114	1.61
(c) 5% anchored	0.31 [0.22, 0.38]	-2.02, 1.23	0.93 [0.77, 1.12]	0.913	0.063	1.44
(d) HN(0, 0.5) prior	0.42 [0.27, 0.75]	-1.65, 0.82	0.86 [0.70, 1.03]	0.883	0.123	1.64
(e) DS initialisation	0.32 [0.26, 0.39]	-2.08, 1.20	0.83 [0.67, 1.00]	0.913	0.084	1.20

In (a) the two entries are $\sigma_{\theta,0} / \sigma_{\theta,1}$. Accuracy and Brier score of the posterior class probabilities against the simulated true classes (all 300 items; for (c) the 15 anchored items are included).

4.4 Results over the grid

Figure 1 shows the bias of prevalence, sensitivity and specificity; Figure 2 the accuracy and Brier score of the posterior labels; Figure 3 the coverage of the CRE-LCM 95% intervals; and Figure 4 the recovery of the variance shares at $N = 300$, $P = 5$. Tables 3–5 give the per-cell values.

Dawid–Skene. The single-data-set pattern of Table 1 holds over the grid. Under conditional independence Dawid–Skene is unbiased for every quantity at every N and P . Its prevalence bias is about +0.003 at $\delta = 0.5$, +0.05 at $\delta = 1$ and +0.10 to +0.11 at $\delta = 1.5$, and its specificity bias +0.04 and +0.09 at $\delta = 1$ and 1.5, essentially independent of N and P : the bias is a property of the misspecification, not of the sample size. Sensitivity is much less affected (at most +0.03). Majority vote is less biased for prevalence (at most +0.05) but has no accuracy estimates.

CRE-LCM with $P = 3$. With three prompts the model removes most of the bias: prevalence bias +0.017, +0.010 and +0.004 at $\delta = 1$ for $N = 150, 300$ and 600 (against +0.05 for Dawid–Skene), specificity bias below 0.02 in absolute value, and coverage of the prevalence interval 0.95 or above at $\delta \leq 1$. At $\delta = 1.5$ the bias is about +0.05 to +0.07, roughly half of Dawid–Skene’s, and coverage falls to 0.73 to 0.98. The Brier score of the posterior labels is lower than Dawid–Skene’s in every cell (for example 0.070 against 0.100 at $\delta = 1$, $N = 300$) and the item, model and prompt shares are recovered with small bias and coverage of 0.80 to 0.95.

CRE-LCM with $P = 5$ at $\delta \geq 1$: a sampler, not an estimator, result. With five prompts and strong dependence the CRE-LCM prevalence bias is not smaller than Dawid–Skene’s (+0.044 to +0.055 at $\delta = 1$; +0.10 to +0.13 at $\delta = 1.5$) and the coverage of its prevalence interval collapses (to 0.02 at $\delta = 1.5$, $N = 600$). The convergence record explains this. In these cells 83% to 97% of replications have a global

parameter with $\text{split-}\hat{R} > 1.05$, the median bulk effective sample size of the worst parameter is 5 to 9 from 1500 draws, the two chains agree with each other (a merged-class mode is detected in only 41 of 2418 classifiable replications), and the CRE-LCM prevalence estimate is within 0.02 of the Dawid–Skene value that initialised the chains and correlates with it at 0.7 to 0.9. The chains have not moved away from their starting point within 1500 iterations. With $P = 3$ the same cells show the estimate drifting away from the initial value towards the truth (-0.04 to -0.05 relative to Dawid–Skene) but still with low effective sample sizes. The identifiability fits of Section 3.4 (four chains of 2000 iterations, the same $\delta = 1$, $N = 300$, $P = 5$ design) gave a prevalence bias of $+0.02$, so the discrepancy is a matter of chain length, and the grid results for CRE-LCM at $\delta \geq 1$ are lower bounds on its performance rather than estimates of it. Section 6 returns to this.

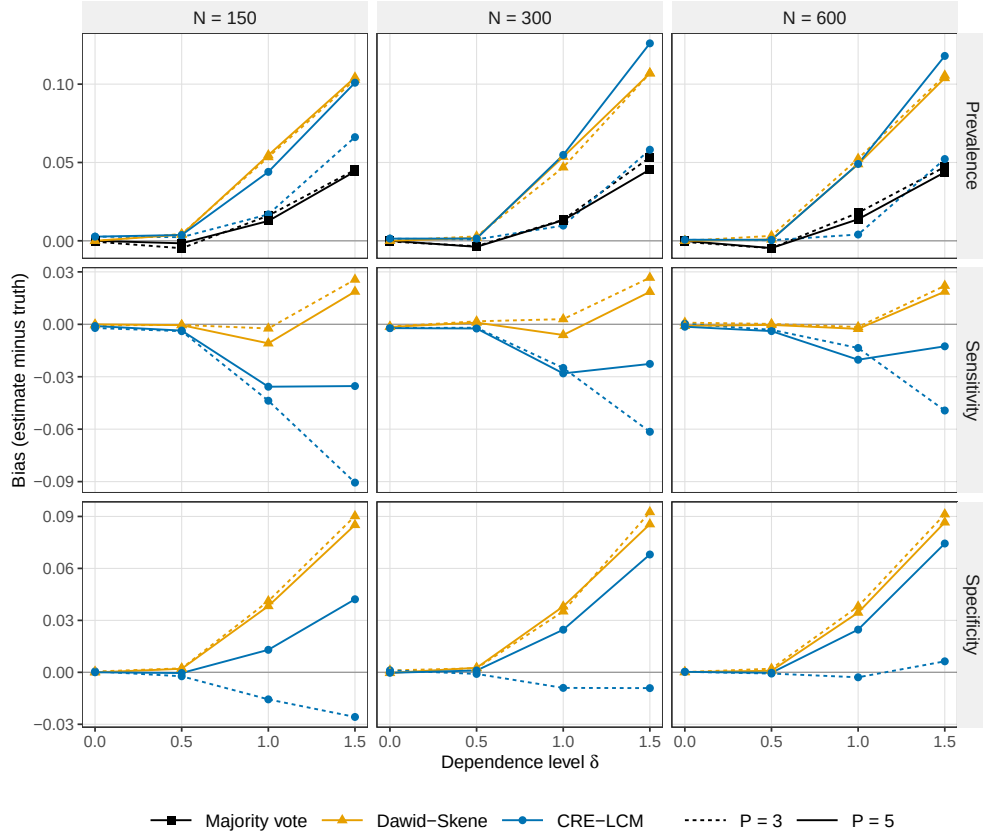


Fig. 1 Bias of the estimated prevalence (top), mean sensitivity (middle) and mean specificity (bottom) by dependence level δ for majority vote, Dawid–Skene and CRE-LCM; columns are sample sizes, line types are the number of prompts. Means over 100 replications per cell

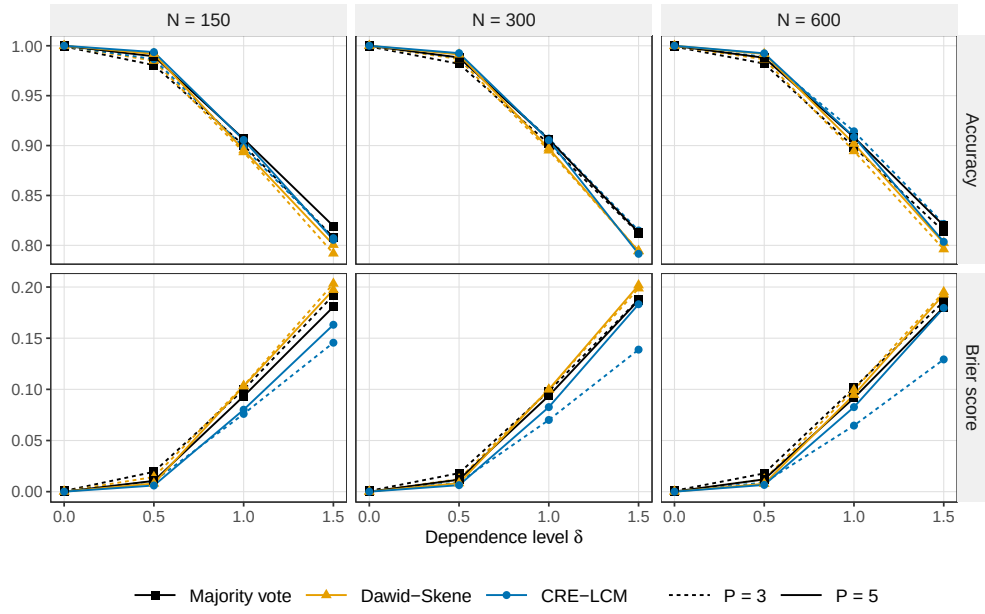


Fig. 2 Classification accuracy (top) and Brier score (bottom) of the posterior class probabilities by dependence level δ ; layout as in Fig. 1

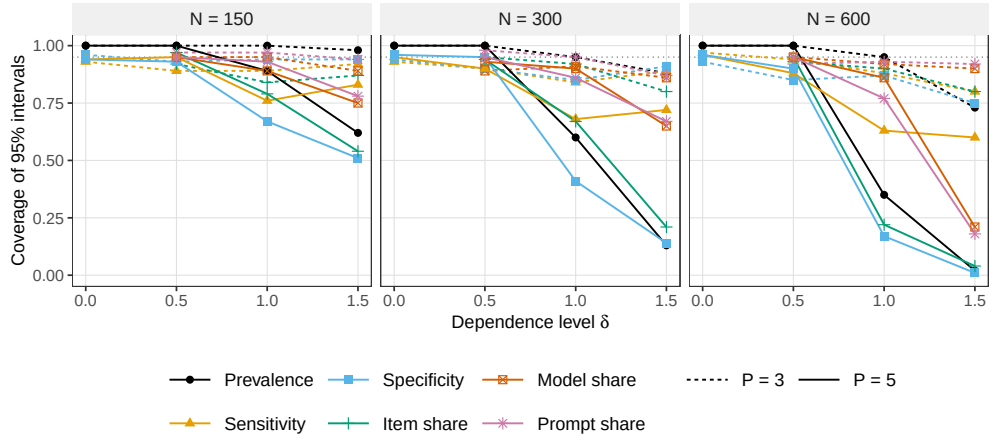


Fig. 3 Coverage of the CRE-LCM 95% credible intervals for prevalence, mean sensitivity, mean specificity and the three variance shares, by dependence level δ and sample size; the dotted line marks the nominal level. Coverage of a share is undefined at $\delta = 0$, where its true value is zero

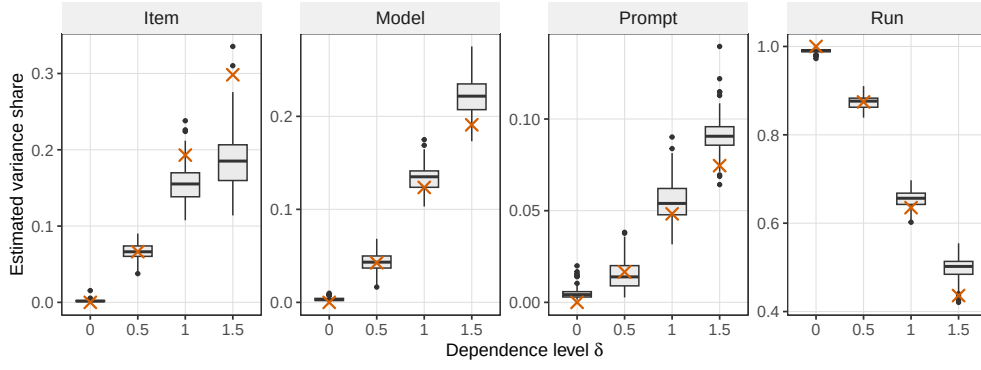


Fig. 4 Posterior means of the four variance shares over 100 replications at $N = 300$, $P = 5$, by dependence level δ ; crosses mark the true shares

Table 3 Prevalence estimation over the simulation grid ($M = 3$, $R = 3$; n replications per cell): bias and root mean squared error of majority vote (MV), Dawid–Skene (DS) and CRE-LCM, and coverage of the CRE-LCM 95% credible interval

P	N	δ	n	MV		DS		CRE-LCM		
				bias	RMSE	bias	RMSE	bias	RMSE	cov.
3	150	0.0	100	-0.001	0.002	0.000	0.001	0.003	0.003	1.00
		0.5	100	-0.005	0.018	0.004	0.011	0.002	0.010	1.00
		1.0	100	0.016	0.043	0.053	0.072	0.017	0.070	1.00
		1.5	100	0.045	0.065	0.103	0.120	0.066	0.104	0.98
	300	0.0	100	0.000	0.002	0.000	0.001	0.001	0.002	1.00
		0.5	100	-0.003	0.014	0.003	0.009	0.001	0.007	1.00
		1.0	100	0.013	0.032	0.047	0.059	0.010	0.049	0.95
		1.5	100	0.053	0.064	0.107	0.116	0.058	0.102	0.88
	600	0.0	100	-0.001	0.002	0.000	0.001	0.001	0.001	1.00
		0.5	100	-0.005	0.012	0.003	0.008	0.000	0.004	1.00
		1.0	100	0.018	0.035	0.052	0.061	0.004	0.028	0.95
		1.5	100	0.047	0.058	0.106	0.112	0.052	0.089	0.73
5	150	0.0	100	0.000	0.000	0.000	0.000	0.003	0.003	1.00
		0.5	100	-0.002	0.009	0.004	0.009	0.004	0.008	1.00
		1.0	100	0.013	0.034	0.055	0.066	0.044	0.065	0.89
		1.5	100	0.044	0.062	0.104	0.117	0.101	0.126	0.62
	300	0.0	100	0.000	0.000	0.000	0.000	0.001	0.001	1.00
		0.5	100	-0.004	0.009	0.002	0.006	0.001	0.006	1.00
		1.0	100	0.013	0.029	0.054	0.063	0.055	0.063	0.60
		1.5	100	0.045	0.053	0.107	0.115	0.126	0.138	0.13
	600	0.0	100	0.000	0.000	0.000	0.000	0.001	0.001	1.00
		0.5	100	-0.005	0.009	0.001	0.004	0.001	0.004	1.00
		1.0	100	0.014	0.025	0.049	0.054	0.049	0.053	0.35
		1.5	100	0.044	0.050	0.104	0.108	0.118	0.125	0.02

Table 4 Bias of mean sensitivity and mean specificity (averaged over the $M \times P$ configurations) for Dawid–Skene (DS) and CRE-LCM, with coverage of the CRE-LCM 95% intervals

P	N	δ	Sensitivity			Specificity		
			DS	CRE	cov.	DS	CRE	cov.
3	150	0.0	-0.001	-0.002	0.93	0.000	0.001	0.96
		0.5	0.000	-0.004	0.89	0.002	-0.002	0.93
		1.0	-0.002	-0.044	0.89	0.041	-0.016	0.94
		1.5	0.026	-0.091	0.92	0.090	-0.026	0.94
		0.0	-0.001	-0.002	0.93	0.001	0.001	0.94
	300	0.5	0.002	-0.002	0.90	0.002	-0.001	0.90
		1.0	0.003	-0.025	0.84	0.035	-0.009	0.85
		1.5	0.027	-0.061	0.89	0.093	-0.009	0.91
		0.0	0.001	0.000	0.97	0.000	0.000	0.93
		0.5	0.000	-0.003	0.94	0.002	-0.001	0.85
	600	1.0	-0.002	-0.014	0.88	0.038	-0.003	0.87
		1.5	0.022	-0.049	0.80	0.091	0.006	0.75
5	150	0.0	0.000	-0.001	0.94	0.000	0.000	0.94
		0.5	0.000	-0.004	0.95	0.002	0.000	0.93
		1.0	-0.011	-0.036	0.76	0.038	0.013	0.67
		1.5	0.019	-0.035	0.83	0.085	0.042	0.51
		0.0	-0.002	-0.002	0.95	0.000	0.000	0.96
	300	0.5	0.001	-0.002	0.90	0.003	0.001	0.95
		1.0	-0.006	-0.028	0.68	0.038	0.025	0.41
		1.5	0.019	-0.023	0.72	0.086	0.068	0.14
		0.0	-0.001	-0.001	0.96	0.000	0.000	0.96
		0.5	0.000	-0.004	0.88	0.001	0.000	0.90
	600	1.0	-0.003	-0.020	0.63	0.034	0.025	0.17
		1.5	0.019	-0.013	0.60	0.087	0.074	0.01

4.5 Misspecification

Table 6 compares the Gaussian and t_4 data-generating processes at $\delta = 1$, $N = 300$, $P = 5$. Heavy-tailed random effects do not degrade CRE-LCM relative to the Gaussian case: prevalence bias +0.034 against +0.055, accuracy 0.929 against 0.905, Brier 0.064 against 0.083, and coverage of the prevalence interval 0.87 against 0.60; Dawid–Skene moves in the same direction (+0.037 against +0.054). The t_4 cell is also less affected by the slow mixing just described (75% flagged against 92%). The misspecification arm therefore shows robustness of the point estimates and labels to the tail behaviour of the random effects at this design size, with the same caveat on chain length.

5 Application: policy-topic coding

[DATA: this section waits on the annotation of Comparative Agendas Project texts; design as planned: $M = 3$ –4 model families including one open-weights model, $P = 5$ semantically equivalent prompts, $R = 3$ runs at temperature above zero, with version strings and dates recorded; human-coded major topics held out from estimation and used as the gold standard for evaluation; variance decomposition, comparison of Dawid–Skene and CRE-LCM against the held-out codes, and posterior predictive checks.]

Table 5 Posterior label quality: classification accuracy and Brier score of the item-level class probabilities (MV and DS: EM posterior; CRE-LCM: posterior mean), the fraction of replications with any split- $\hat{R} > 1.05$ among the CRE-LCM global parameters, and mean CRE-LCM runtime per replication (minutes, two chains of 1500 iterations on one core)

P	N	δ	Accuracy			Brier			$\hat{R} > 1.05$	min
			MV	DS	CRE	MV	DS	CRE		
3	150	0.0	0.999	1.000	1.000	0.001	0.000	0.000	0.00	1.1
		0.5	0.981	0.985	0.987	0.019	0.014	0.010	0.05	1.1
		1.0	0.900	0.893	0.899	0.100	0.103	0.076	0.94	1.4
		1.5	0.808	0.792	0.807	0.192	0.203	0.146	0.89	1.8
	300	0.0	0.999	1.000	1.000	0.001	0.000	0.000	0.01	2.7
		0.5	0.982	0.987	0.988	0.018	0.012	0.009	0.13	2.7
		1.0	0.902	0.895	0.907	0.098	0.100	0.070	0.95	2.9
		1.5	0.812	0.795	0.815	0.188	0.199	0.139	0.99	3.7
	600	0.0	0.999	1.000	1.000	0.001	0.000	0.000	0.02	7.5
		0.5	0.982	0.986	0.989	0.018	0.012	0.009	0.17	7.5
		1.0	0.899	0.894	0.914	0.101	0.099	0.065	0.99	7.5
		1.5	0.814	0.796	0.822	0.186	0.195	0.129	0.98	8.5
5	150	0.0	1.000	1.000	1.000	0.000	0.000	0.000	0.00	1.8
		0.5	0.989	0.992	0.994	0.011	0.008	0.006	0.18	1.8
		1.0	0.907	0.895	0.906	0.093	0.103	0.080	0.83	2.0
		1.5	0.819	0.801	0.806	0.181	0.197	0.163	0.92	2.1
	300	0.0	1.000	1.000	1.000	0.000	0.000	0.000	0.01	4.4
		0.5	0.988	0.991	0.993	0.012	0.008	0.006	0.19	4.6
		1.0	0.906	0.897	0.905	0.094	0.100	0.083	0.92	4.6
		1.5	0.813	0.794	0.792	0.187	0.202	0.183	0.97	4.8
	600	0.0	1.000	1.000	1.000	0.000	0.000	0.000	0.03	11.9
		0.5	0.988	0.992	0.992	0.012	0.008	0.007	0.20	12.2
		1.0	0.908	0.902	0.909	0.092	0.095	0.083	0.92	11.8
		1.5	0.820	0.802	0.804	0.180	0.193	0.180	0.97	12.8

Table 6 Recovery of the variance shares by CRE-LCM at $N = 300$, $P = 5$ (bias, RMSE and 95% coverage), with the misspecification arm in which the three item random effects follow a scaled t_4 distribution at $\delta = 1$; the last three columns give the CRE-LCM prevalence bias, accuracy and Brier score in the same cells

DGP	δ	n	Item share			Model share			Prompt share			CRE-LCM		
			bias	RMSE	cov.	bias	RMSE	cov.	bias	RMSE	cov.	prev.	bias	acc. Brier
Gaussian	0.0	100	0.002	0.003	—	0.003	0.004	—	0.005	0.006	—	0.001	1.000	0.000
Gaussian	0.5	100	0.000	0.011	0.94	0.001	0.010	0.93	-0.001	0.008	0.95	0.001	0.993	0.006
Gaussian	1.0	100	-0.037	0.044	0.67	0.010	0.017	0.90	0.007	0.012	0.86	0.055	0.905	0.083
Gaussian	1.5	100	-0.110	0.117	0.21	0.031	0.038	0.65	0.017	0.020	0.67	0.126	0.792	0.183
t_4	1.0	100	-0.067	0.071	0.14	0.002	0.015	0.93	0.007	0.013	0.85	0.034	0.929	0.064

6 Discussion

What the variance decomposition is for. The three random-effect shares answer a question that agreement statistics cannot: where should the next unit of annotation budget go? Decoding runs only reduce the residual share; they cannot touch item, model or prompt variance. When the item-by-model share dominates, a fourth model

family is worth more than any number of re-runs; when the item-by-prompt share dominates, the prompt family should be widened. The shared item share is the ceiling: it is the part of disagreement that reflects genuine ambiguity in the text and is reduced only by better coding rules or human adjudication. We therefore recommend that studies using LLM annotation report the three shares with their intervals next to the customary κ , and that “temperature 0, single prompt, single model” be treated as a design with no replication rather than as a deterministic instrument.

Relation to bounds and to design-based correction. [Chen et al. \(2026\)](#) show that if the dependence among LLM annotators is left unrestricted, only bounds on prevalence are available. Our results locate the price of that generality: with the dependence structured by the annotation design, point estimates are recovered, at the cost of an assumption (crossed Gaussian effects) that we check by misspecification. Design-based methods ([Egami et al. 2024](#); [Angelopoulos et al. 2023](#)) reach validity from the opposite direction, by spending a labelled subsample on a doubly robust correction; our anchoring variant shows that a subsample of the same order also removes the estimation difficulties of the latent class model, and the two approaches are complementary rather than competing—one supplies a measurement model, the other a downstream-inference correction.

Accuracy is not enough. [Carlson and Burbano \(2026\)](#) found that a configuration with 88% first-stage accuracy produced a downstream coefficient of the wrong sign while one with 56% accuracy did not, because the direction of error was correlated with the outcome. Our model does not solve that problem, but it makes its ingredients visible: the class-specific model effects a_{mk} show which configurations err asymmetrically, and the posterior class probabilities carry the residual uncertainty into any downstream analysis instead of discarding it.

Limitations. The main model is binary; the multiclass extension is sketched and left for future work. The random effects are Gaussian and crossed; our misspecification arm shows [SIM] but other departures (bimodal ambiguity, model-by-prompt interactions) are untested. The shared item scale is shrunk by 10–20% at $N = 300$ in the correct mode, which propagates into the item share; anchoring reduces the shrinkage and larger N [SIM]. Estimation by MCMC takes minutes per fit; an EM algorithm with quadrature would make the method routine. Finally, LLM versions change: the annotations we report are tied to the version strings and dates in the supplementary material, and the model’s parameters should be understood as properties of those versions, not of the model families.

7 Conclusion

An LLM annotation ensemble is a designed experiment with three crossed factors, and its errors inherit that design. Treating each configuration as an independent rater, as Dawid–Skene does, overstates how good the configurations are and, as dependence grows, biases what they measure. Giving the latent class model the structure of the design restores point identification without human labels, decomposes annotation noise into components that map onto concrete design decisions, and produces class probabilities that are better calibrated than majority vote or Dawid–Skene. The main practical

obstacles—a merged-class posterior mode and slow mixing on the prevalence—are addressed by initialising at the Dawid–Skene solution or by anchoring a small labelled subset. Code, prompts, model versions and all annotations are provided for reuse.

Supplementary information. Stan code for all model variants, additional tables and the full per-cell simulation results are provided as Online Resource 1. The code repository is available through an anonymised link for review.

Appendix A Stan code

```
// crossed_lcre.stan
// Binary latent class model with crossed random effects for
// LLM annotation ensembles (model x prompt x run).
// The discrete class c_i is marginalised out.

data {
  int<lower=1> N; // items
  int<lower=1> M; // LLM models
  int<lower=1> P; // prompts
  int<lower=1> R; // runs per (model, prompt)
  array[N, M, P] int<lower=0, upper=R> S; // # runs labelled 1
}

parameters {
  real<lower=0, upper=1> pi1; // prevalence of class 1
  ordered[2] mu; // class intercepts (label identification)
  matrix[M - 1, 2] a_raw; // model effects, model 1 = reference
  matrix[P, 2] b_z; // prompt effects (non-centred)
  vector<lower=0>[2] tau_b;
  real<lower=0> s_th; // item ambiguity SD
  real<lower=0> s_ph; // item x model SD
  real<lower=0> s_ps; // item x prompt SD
  vector[N] th_z;
  matrix[N, M] ph_z;
  matrix[N, P] ps_z;
}

transformed parameters {
  matrix[M, 2] a;
  matrix[P, 2] b;
  a[1] = rep_row_vector(0, 2);
  if (M > 1) a[2:M] = a_raw;
  for (k in 1:2) b[, k] = tau_b[k] * b_z[, k];
}

model {
  // priors
  pi1 ~ beta(1, 1);
  mu ~ normal(0, 2);
  to_vector(a_raw) ~ normal(0, 1);
  to_vector(b_z) ~ std_normal();
  tau_b ~ normal(0, 0.5);
  s_th ~ normal(0, 1);
  s_ph ~ normal(0, 1);
  s_ps ~ normal(0, 1);
  th_z ~ std_normal();
  to_vector(ph_z) ~ std_normal();
  to_vector(ps_z) ~ std_normal();

  // likelihood, marginalising the latent class
  for (i in 1:N) {
    vector[2] lp = [log1m(pi1), log(pi1)]';
    for (m in 1:M) {
      for (p in 1:P) {

```

```

        real u = s_th * th_z[i] + s_ph * ph_z[i, m] + s_ps * ps_z[i, p];
        for (k in 1:2)
            lp[k] += binomial_logit_lpmf(S[i, m, p] | R, mu[k] + a[m, k] + b[p, k] + u);
    }
    target += log_sum_exp(lp);
}
}

generated quantities {
    vector[N] post1; // P(c_i = 1 | data)
    vector[N] log_lik; // pointwise (per item) log-likelihood for L00
    // variance shares on the latent logistic scale (run-level residual = pi^2/3)
    real v_res = square(pi()) / 3;
    real v_tot = square(s_th) + square(s_ph) + square(s_ps) + v_res;
    real share_item = square(s_th) / v_tot;
    real share_model = square(s_ph) / v_tot;
    real share_prompt = square(s_ps) / v_tot;
    real share_run = v_res / v_tot;
    for (i in 1:N) {
        vector[2] lp = [log1m(pi1), log(pi1)]';
        for (m in 1:M) for (p in 1:P) {
            real u = s_th * th_z[i] + s_ph * ph_z[i, m] + s_ps * ps_z[i, p];
            for (k in 1:2)
                lp[k] += binomial_logit_lpmf(S[i, m, p] | R, mu[k] + a[m, k] + b[p, k] + u);
        }
        log_lik[i] = log_sum_exp(lp);
        post1[i] = exp(lp[2] - log_lik[i]);
    }
}

```

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